Spatial–Spectral Bipartite Graph Clustering With Low-Frequency Tensor Regularization for Hyperspectral and LiDAR Data

Zhe Cao, Yihang Lu, Haonan Xin, Chuanqiang Yu, Rong Wang, and Feiping Nie

***Abstract*—The increasing complexity of remote sensing (RS) applications necessitates multimodal data fusion to overcome the inherent limitations of single-source data. In particular, the integration of hyperspectral images (HSIs) and light detection and ranging (LiDAR) data captures complementary spectral and spatial features, significantly enhancing object discrimination. Unlike supervised methods requiring costly expert annotations, unsupervised clustering eliminates labeling needs, thereby offering an efficient solution for complex scene analysis with reduced deployment costs. However, existing unsupervised clustering methods still face several challenges, including insufficient utilization of spatial information, high data dimensionality, and interference from high-frequency noise, which collectively hinder their further development and practical application. To address these issues, this article proposes a novel method called spatial–spectral bipartite graph clustering with low-frequency tensor (SSBC-LFT) regularization for hyperspectral and LiDAR data. The proposed method first focuses on modeling clustering structures, innovatively constructing spatially informed bipartite graphs for each modality to enrich spatial details while reducing computational complexity. Additionally, an adaptive spectral band selection mechanism dynamically assigns higher weights to discriminative bands during clustering while suppressing redundant ones. Finally, through proposed low-frequency tensor nuclear norm (LFTNN) regularization, high-frequency noise is effectively filtered in the frequency domain, extracting stable and consistent cross-modal structural information from the low-rank space to further enhance clustering robustness and accuracy. Extensive experiments on real-world multimodal RS datasets demonstrate the superior performance of the SSBC-LFT model, offering a robust and effective solution for unsupervised clustering in complex scenarios.**

***Index Terms*—Band selection, low-frequency tensor nuclear norm (LFTNN), multimodal remote sensing (RS), spatial bipartite graph.**

I. Introduction

REMOTE sensing (RS) [1], [2], [3] data has gained substantial attention in recent years due to its broad applications in various fields, including environmental monitoring, land-use classification, and agricultural analysis [4], [5], [6]. The development and integration of multimodal RS data, such as hyperspectral images (HSIs), light detection and ranging (LiDAR), and synthetic aperture radar (SAR), have provided researchers with unprecedented opportunities to capture comprehensive information about the Earth's surface [7], [8]. HSIs with their high spectral resolution allow for detailed identification of materials based on their spectral signatures [9], [10], while LiDAR provides accurate elevation data, and SAR offers insights into surface roughness and moisture [11]. The fusion of these diverse data sources enables a more holistic analysis, improving the accuracy and depth of surface characterization tasks [12]. However, challenges such as computational complexity, redundancy, and noise in multimodal data integration have hindered the potential of these techniques [13], [14], [15], [16].

Traditional remote methods, whether for classification or clustering tasks, tend to focus on single-modality data, often leading to incomplete or suboptimal results. For instance, HSIs alone may provide detailed spectral information but often suffer from spectral variability, where similar materials may exhibit different spectral signatures due to environmental factors such as lighting conditions or atmospheric effects [17]. LiDAR, while excellent for capturing topographical details, lacks the material-specific information that HSIs provide [18]. Consequently, the integration of multimodal data has emerged as a key research focus, aiming to leverage the complementary strengths of each modality to enhance the accuracy of recognition and analysis tasks [19], [20], [21], [22], [23].

Multimodal data analysis typically involves two major approaches: classification and clustering [24], [25]. In supervised classification, models are trained on labeled data to learn discriminative features for various classes. In recent years, deep learning methods, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and, more recently, Transformer models, have demonstrated significant improvements in classification tasks by fusing features from multiple data sources [26]. Graph-based neural networks, particularly graph convolutional networks (GCNs), have also been employed to model the relationships between pixels or regions, offering enhanced capabilities in capturing complex spatial and spectral relationships [27]. While these supervised methods have proved effective, they rely heavily on the

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availability of large amounts of labeled training data [28]. This dependence poses a significant limitation, particularly in RS applications where acquiring labeled data is labor-intensive, costly, and often impractical over large geographic areas [29].

Unsupervised learning methods, particularly clustering, provide a promising alternative by eliminating the need for labeled data [30]. Clustering aims to group similar pixels or regions based on their inherent characteristics, allowing for automatic categorization of land-cover types or other surface features [31]. Graph-based methods have become a popular choice for unsupervised clustering, as they can model the relationships between data points through weighted affinity matrices, which represent the similarity between pixels or regions [32]. Despite the promise of these methods, they face several challenges. One of the most significant is the underutilization of spatial information. In RS data, spatially adjacent pixels are often highly correlated, yet many graph-based methods fail to incorporate this spatial context, leading to suboptimal clustering results [33], [34]. Additionally, the high computational cost of constructing and processing affinity matrices, particularly for large-scale datasets, poses a substantial challenge [35].

One of the critical challenges in unsupervised clustering of hyperspectral data is the presence of redundant or noisy spectral bands [36], particularly in HSIs. These redundant spectral bands not only significantly increase storage and computational costs but also exacerbate data transmission bandwidth requirements, thereby reducing the model's real-time performance and computational efficiency, especially in large-scale hyperspectral clustering tasks [37]. Furthermore, the presence of redundant spectral bands intensifies the "curse of dimensionality" in high-dimensional data, diminishing feature discriminability and adversely affecting the stability and accuracy of clustering algorithms [38]. In addition, the sensitivity of the optical system to the chief ray angle (CRA) can induce co-channel interference (CCI), a phenomenon that may be further exacerbated in the presence of excessive redundant spectral bands [39]. When CRA variations are significant, the pixel response locations on the detector may shift [40], leading to spectral misalignment and crosstalk between adjacent channels, thereby compromising spectral independence and reducing spectral resolution. Additionally, signal diffusion across multiple channels may further result in a decline in the signal-to-noise ratio (SNR) [41], further degrading the overall quality of hyperspectral data and negatively impacting subsequent clustering and classification tasks.

More importantly, hyperspectral data is highly susceptible to high-frequency noise, which not only introduces additional spectral distortions but also obscures the underlying low-rank structures that are critical for clustering tasks [42], [43], [44], [45]. However, existing methodologies often exhibit significant limitations in addressing these challenges, particularly in multimodal RS applications, such as the fusion of hyperspectral, LiDAR, and other RS data. In such scenarios, these issues become even more pronounced, directly compromising the stability and generalization capability of clustering algorithms.

In response to these challenges, we propose a novel approach, named spatial–spectral bipartite graph clustering with low-frequency tensor (SSBC-LFT) regularization for hyperspectral and LiDAR data, which introduces several key innovations aimed at addressing the limitations of current methods. The main contributions of our approach are summarized as follows.

1) *Spatially Enriched Bipartite Graph Construction:* One of the significant limitations of traditional graph-based clustering methods is their failure to incorporate spatial information effectively, which is essential in RS applications. Our method addresses this by proposing an innovative bipartite graph construction that integrates spatial information directly into the graph model. This spatial enrichment reduces the computational complexity of graph construction by decreasing the number of anchors, while preserving the richness of spatial information and ensuring that spatially adjacent pixels, which typically belong to the same class, are grouped accordingly.
2) *Adaptive Spectral Band Selection:* Hyperspectral data is characterized by high dimensionality, with many spectral bands often being redundant or irrelevant for specific tasks. To overcome this issue, we propose an adaptive spectral band selection mechanism that assigns greater weights to important spectral bands while reducing the influence of less relevant ones. This method reduces computational costs while preserving critical information.
3) *Low-Frequency Tensor Nuclear Norm (LFTNN):* High-frequency noise in hyperspectral data often obscures the low-rank structures critical for accurate clustering, especially in multimodal contexts where data heterogeneity exacerbates inconsistencies. We propose the LFTNN, a pioneering framework designed to tackle this challenge. Unlike conventional norms, the LFTNN focuses on low-frequency components to suppress high-frequency noise and retain robust structural information. This innovative norm represents a significant advancement, offering a transformative solution for multimodal RS tasks.

In the following sections, we will systematically present the theoretical foundations and key design components of the proposed SSBC-LFT model. In Section IV, we will sequentially introduce the extraction of spatial information [Section (IV-A)], the adaptive spectral band selection mechanism [Section (IV-B)], and the LFTNN regularization strategy [Section (IV-C)]. Section V presents extensive experiments on three real-world multimodal RS datasets. Section VI discusses the broader implications of SSBC-LFT for unsupervised learning, and Section VII outlines potential future directions.

II. Notations

In this article, we denote third-order tensors using bold calligraphic letters, for example, $\mathcal{A}$, matrices with bold uppercase letters, for example, $\mathbf{A}$, vectors with bold lowercase letters, for example, $\mathbf{a}$, and scalars are denoted by lowercase, nonbold letters, for example, a. The fields of real numbers are denoted as $\mathbb{R}$. For a matrix $\mathbf{A} \in \mathbb{R}^{n_1 \times n_2}$, $\mathbf{A}^T$ denotes the transpose of $\mathbf{A}$, and $tr(\mathbf{A})$ denotes its matrix trace. We represent $\mathbf{I} \in \mathbb{R}^{n_3 \times n_3}$ as the identity matrix and $\mathbf{1} \in \mathbb{R}^{n_3 \times 1}$ as the matrix that is entirely composed of ones. Here, $n_1, n_2, n_3 \in \mathbb{N}^+$ denote the matrix dimensions. Regarding a third-order tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, we

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denote its (i, j, k)th entry as $\mathcal{A}_{ijk}$ and the tube is designated as $\mathcal{A}(i, j, :)$. We utilize the MATLAB notation $\mathcal{A}(k, :, :)$, $\mathcal{A}(:, k, :)$, and $\mathcal{A}(:, :, k)$ to denote, respectively, the kth horizontal, lateral, and frontal slice [12]. In numerous instances, the kth frontal slice denoted by $\mathcal{A}(:, :, k)$ can be more succinctly represented as A^k. Additionally, $\tilde{\mathcal{A}} = fft(\mathcal{A}, [], 3)$ denotes the fast Fourier transform (FFT) [13] applied along the third dimension of $\mathcal{A}$. Conversely, we use the inverse FFT (IFFT) [13] to revert $\tilde{\mathcal{A}}$ to $\mathcal{A}$, which is expressed as $\mathcal{A} = ifft(\tilde{\mathcal{A}}, [], 3)$.

III. Preliminaries

A. T-Product and T-SVD

Definition 1 (T-product [32]): Let $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\mathcal{B} \in \mathbb{R}^{n_2 \times l \times n_3}$. Then, the T-product $\mathcal{A} * \mathcal{B}$ is defined to be a tensor $\mathcal{C}$ of size $n_1 \times l \times n_3$ as

$$\mathcal{C} = \mathcal{A} * \mathcal{B} = fold\left(bcirc\left(\mathcal{A}\right) \cdot unfold\left(\mathcal{B}\right)\right) \tag{1}$$

where

$$bcirc\left(\mathcal{A}\right) = \begin{bmatrix} A^1 & A^{n_3} & \cdots & A^2 \\ A^2 & A^1 & \cdots & A^3 \\ \vdots & \vdots & \ddots & \vdots \\ A^{n_3} & A^{n_3-1} & \cdots & A^1 \end{bmatrix} \tag{2}$$

$$unfold\left(\mathcal{A}\right) = \begin{bmatrix} A^1 \\ A^2 \\ \vdots \\ A^{n_3} \end{bmatrix}, \quad fold\left(unfold\left(\mathcal{A}\right)\right) = \mathcal{A}. \tag{3}$$

The T-product is comparable to matrix multiplication in the Fourier domain. Specifically, $\mathcal{C} = \mathcal{A} * \mathcal{B}$ can be viewed as $\tilde{\mathcal{C}} = \tilde{\mathcal{A}}\tilde{\mathcal{B}}$. This characteristic indicates the possibility of using the FFT-based approach to calculate the T-product, rather than relying on formula (1).

Definition 2 (f-Diagonal Tensor [32]): A tensor is referred to as frontal-diagonal if each of its frontal slices consistently manifests as a diagonal matrix.

Definition 3 (Tensor Transpose [32]): The transpose of a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is the tensor $\mathcal{A}^T \in \mathbb{R}^{n_2 \times n_1 \times n_3}$ obtained by transposing each of the frontal slices and then reversing the order of transposed frontal slices 2 through n_3.

As an example, let $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times 4}$ and its frontal slices be A^1, A^2, A^3, and A^4. Then

$$\mathcal{A}^T = fold\left(\begin{bmatrix} A^{1T} \\ A^{4T} \\ A^{3T} \\ A^{2T} \end{bmatrix}\right). \tag{4}$$

Definition 4 (Tensor Frobenius norm [32]): Given a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its tensor Frobenius norm is defined as

$$\|\mathcal{A}\|_F = \sqrt{\sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \sum_{k=1}^{n_3} \left|\mathcal{A}_{ijk}\right|^2}. \tag{5}$$

Definition 5 (T-SVD [46]): Given $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, it is possible to factorize it as follows:

$$\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T \tag{6}$$

where $\mathcal{U} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are orthogonal tensors, $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is a tensor with an f-diagonal structure, and $\mathcal{V}^T$ is the transpose of $\mathcal{V}$.

IV. Proposed Method

Clustering is an unsupervised learning technique that groups similar objects or data points without the need for GT labels or expert annotations. We denote pixels in an HSI as a set $\mathbf{X}^1 = \{x_i^1\}_{i=1}^N \in \mathbb{R}^{D_1}$, where each x_i^1 denotes the spectral signature of a pixel in the image, D_1 represents the number of spectral bands in the HSI, and N is the total number of pixels. We denote pixels in an LiDAR as a set $\mathbf{X}^2 = \{x_i^2\}_{i=1}^N \in \mathbb{R}^{D_2}$, where each $\{x_i^2\}$ denotes the spectral signature of a pixel in the image, and D_2 represents the number of spectral bands. HSI clustering algorithms partition pixels into a clustering $\{C_k\}_{k=1}^K$, where each C_k represents a cluster and K denotes the total number of clusters. The pixels within the same cluster are considered "similar" (possibly due to a similar material constitution), while pixels from different clusters are "dissimilar" (containing different materials). The specific notion of similarity used varies widely across the many clustering algorithms in the literature.

To understand the key advancements of the proposed SSBC-LFT enriched with spatial information for hyperspectral and LiDAR Data, it is essential to explore its four pivotal aspects: spatiality, efficiency, consistency, and robustness. Each of these components addresses critical challenges in RS data processing, ensuring improved performance.

1) *Spatial-Context Integration for Precision:* The SSBC-LFT framework leverages entropy rate superpixel (ERS) segmentation to integrate spatial context with spectral data, addressing the common challenge of spectral variability in RS. Unlike traditional clustering that overlooks spatial relationships, SSBC-LFT harnesses spatial details from high-resolution imagery, using contextual information from neighboring pixels that often belong to the same class and exhibit similar spectral features. This spatial enhancement is crucial in reducing noise and yielding consistent clustering results, as spatially adjacent pixels typically share homogeneous characteristics. Compared to existing methods, SSBC-LFT selects fewer yet strategically representative anchor points, optimizing computational resources while achieving high clustering accuracy.
2) *Adaptive Spectral Band Selection for Efficiency:* The adaptive spectral band selection mechanism within SSBC-LFT combines spectral selection with clustering, optimizing data quality and efficiency in hyperspectral analysis. This method assigns greater weight to the most informative spectral bands, reducing interference from less relevant bands, which lowers computational costs and emphasizes key features. Focusing on the most informative spectral bands significantly enhances clustering performance and robustness, especially for complex, high-dimensional data. This adaptability enables efficient processing and interpretation, enhancing practicality for real-world applications.

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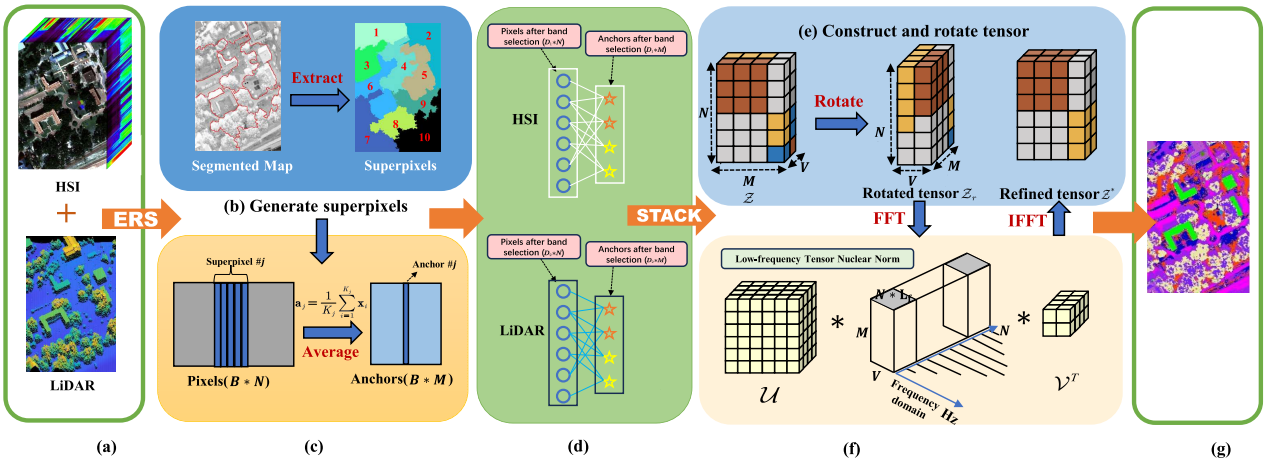


Fig. 1. SSBC-LFT integrates hyperspectral and LiDAR data in the form of spatial bipartite graphs, adaptive band selection, and LFTNN for enhanced clustering. (a) Multimodal data. (b) Generate superpixels. (c) Generate anchor points. (d) Generate bipartite graphs. (e) Construct and rotate a tensor. (f) LFTNN. (g) Clustering results.

3) *Multimodal Consistency for Cohesive Integration:* SSBC-LFT achieves comprehensive data integration by stacking multimodal data into tensors, allowing for thorough examination of consistent structures within a low-rank space across modalities. In RS, combining data from various sources (e.g., HSI, LiDAR, and SAR) provides a complete perspective, but can introduce inconsistencies. The SSBC-LFT model effectively consolidates complementary information, reducing divergence and enhancing clustering stability and accuracy. This method is particularly advantageous for applications like land cover classification, where multimodal consistency improves overall reliability.
4) *Noise Robustness via Low-Frequency Filtering:* To address high-frequency noise, SSBC-LFT introduces an LFTNN technique, isolating consistent, low-rank data while filtering out unwanted information. Traditional clustering often struggles with high-dimensional RS data, where noise can obscure valuable information. This low-frequency filtering focuses on extracting coherent information, significantly reducing noise interference, and enhancing clustering resilience. The method ensures reliable clustering results, making it suitable for large-scale applications where noise reduction is essential.

A. Spatially Informed Anchor Generation Strategy

Traditional graph-based clustering often neglects hyperspectral data's spatial context and struggles with large-scale datasets. We propose an anchor graph-based method integrating ERS segmentation to select spatially representative anchor points. This approach reduces computational complexity by constructing a reduced-dimensional bipartite graph, capturing spatial structures effectively while ensuring efficient processing of extensive datasets.

1) Joint Superpixel Segmentation Process: In HSIs, pixels sharing similar labels are typically grouped within homogeneous areas. Studies have shown that superpixel segmentation can reveal the intrinsic spectral–spatial structure of HSIs, offering a foundation for further analysis. Among the various segmentation methods available, the ERS segmentation approach is adopted here due to its adaptability in generating homogeneous regions of varying shapes and sizes, based on complex spatial and textural patterns. ERS, initially created for RGB images, involves converting the RGB format to grayscale before segmentation. However, since HSIs contain multiple spectral bands, a representative component must first be selected for segmentation. In this case, we use the first principal component, which captures the essential information from HSIs. Formally, let HSI be denoted as a third-order tensor $\boldsymbol{\mathcal{X}}^1 \in \mathbb{R}^{W\times H\times D_1}$, where $W \times H$ represents the spatial dimensions and D_1 the spectral bands. To further enrich the segmentation process, the elevation data from LiDAR, structured as a tensor $\boldsymbol{\mathcal{X}}^2 \in \mathbb{R}^{W\times H\times D_2}$, is integrated with HSIs for joint segmentation. This joint segmentation fully leverages the complementary characteristics of the two modalities: the dense spectral information of HSIs compensates for the inherent sparsity of LiDAR data, while the geometric and spatial structures provided by LiDAR refine the often imprecise spatial boundaries of HSIs caused by mixed pixels and spectral redundancy. As illustrated in Fig. 1(a), the fused data are subsequently concatenated and reshaped into a matrix $\boldsymbol{E} \in \mathbb{R}^{(D_1+D_2)\times N}$, where $N = W \times H$. For efficient processing, principal component analysis (PCA) [47] is initially applied to $\boldsymbol{E}$, after which the first principal component I_p is segmented using ERS, generating M homogeneous superpixels as follows:

$$I_p = \bigcup_{i=1}^{M} \mathcal{S}_i, \text{ s.t. } \mathcal{S}_i \cap \mathcal{S}_j = \varnothing, \quad (i \neq j) \tag{7}$$

where $\mathcal{S}_i$ is the ith local region with the same segmentation label. The constraint $\mathcal{S}_i \cap \mathcal{S}_j = \varnothing$ ensures that any two generated local regions are nonoverlapping. Moreover, all modalities share this unified segmentation result, meaning that they are aligned on the same underlying spatial structure.

2) Spatial Anchor Generation: In anchor-based clustering approaches, the key to efficient data representation lies in selecting a minimal yet informative set of anchor points to

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capture the overall sample distribution. Conventional methods often struggle to choose high-quality anchors for HSI and LiDAR data, primarily because they overlook the spatial organization of pixels. To address this challenge, we propose a spatial anchor generation method that identifies highly informative anchors based on the joint superpixel segmentation of HSI and LiDAR data. Each spatial anchor is derived from a continuous and homogeneous spatial region, thereby fully integrating spatial contextual information into the feature representation process. Since all modalities share the same superpixel partition, the number of anchors generated for each modality remains consistent, which significantly enhances cross-modal structural consistency and spatial alignment. Unlike traditional strategies, our approach selects fewer anchors with increased representativeness, yielding enhanced clustering results. Specifically, for each modality, the anchor matrix, represented as $\boldsymbol{A}^v = \{\boldsymbol{a}_1^v, \boldsymbol{a}_2^v, \ldots, \boldsymbol{a}_M^v\} \in \mathbb{R}^{D_v \times M}$, consolidates M anchors. As illustrated in Fig. 1(c), each anchor is calculated by averaging all pixels within a superpixel, formulated as

$$\boldsymbol{a}_j^v = \frac{1}{N_j} \sum_{i=1}^{N_j} \boldsymbol{x}_{ij}^v, \quad j = 1, 2, \ldots, M \tag{8}$$

where $\boldsymbol{x}_{ij}^v$ represents the ith pixel in the jth superpixel of the vth mode and N_j denotes the number of pixels in the jth superpixel. Therefore, $N = \sum_{j=1}^{M} N_j$. This averaging strategy enables each anchor to capture the aggregated statistical characteristics of its corresponding region, rather than relying on potentially noisy or sparsely distributed individual pixels.

B. Learning Structured Bipartite Graphs

Bipartite graph-based clustering is an effective approach for partitioning samples into clusters, offering significant efficiency advantages over traditional graph clustering methods, which typically require $\mathcal{O}(N^2)$ or $\mathcal{O}(N^3)$ computational complexity. By contrast, bipartite graph clustering operates with a reduced time complexity of $\mathcal{O}(N)$, making it particularly suitable for large-scale, multimodal RS data. To construct the bipartite graph $\boldsymbol{Z}^v \in \mathbb{R}^{N \times M}$ for multimodal RS data, as shown in Fig. 1(d), each pixel $\boldsymbol{x}_i^v \in \boldsymbol{X}^v$ is linked to an anchor point $\boldsymbol{a}_j^v \in \boldsymbol{A}^v$ with an association probability Z_{ij}^v that represents their "pixel-to-anchor" relationship. In this structure, the entry Z_{ij}^v within $\boldsymbol{Z}^v$ quantifies the likelihood of a connection between $\boldsymbol{x}_i^v$ and $\boldsymbol{a}_j^v$, where higher values of Z_{ij}^v indicate a closer proximity between the pixel and anchor point. Consequently, bipartite graph learning for $\boldsymbol{Z}^v$ is achieved through an adaptive assignment of pixels to their nearest anchors, formulated as follows:

$$\begin{aligned} &\min_{\boldsymbol{Z}^v} \sum_{v=1}^{V} \sum_{i=1}^{N} \sum_{j=1}^{M} \left\| \boldsymbol{x}_i^v - \boldsymbol{a}_j^v \right\|_2^2 Z_{ij}^v \\ &\text{s.t. } \boldsymbol{Z}^v \mathbf{1} = \mathbf{1}, \ 0 \le Z_{ij}^v \le 1 \end{aligned} \tag{9}$$

where $\|\boldsymbol{x}_i^v - \boldsymbol{a}_j^v\|_2^2$ represents the squared Euclidean distance between $\boldsymbol{x}_i^v$ and $\boldsymbol{a}_j^v$.

One of the main challenges in unsupervised clustering of multimodal RS data lies in dealing with redundant or noisy spectral bands. For instance, HSIs typically consist of hundreds of narrow spectral bands, many of which are redundant or irrelevant for specific clustering tasks. These unnecessary bands not only increase the dimensionality of the data, resulting in higher computational costs, but also introduce noise that disrupts the clustering process and ultimately lowers clustering accuracy. To address these issues, we incorporate an adaptive spectral band selection mechanism into the clustering process. Specifically, we introduce a diagonal matrix $\boldsymbol{\theta}$ for each modality in bipartite graph learning, where $\boldsymbol{\Theta}^v = diag(\boldsymbol{\theta})$. This mechanism adaptively adjusts the weight θ_d of each spectral band during clustering, assigning higher weights to the most informative bands while reducing the impact of less relevant bands, as

$$\begin{aligned} &\min_{\boldsymbol{\Theta}^v, \boldsymbol{\mathcal{Z}}} \sum_{v=1}^{V} \sum_{i=1}^{N} \sum_{j=1}^{M} \left\| \boldsymbol{\Theta}^v x_i^v - \boldsymbol{\Theta}^v a_j^v \right\|_2^2 Z_{ij}^v \\ &\text{s.t. } \boldsymbol{Z}^v \mathbf{1} = \mathbf{1}, 0 \le Z_{ij} \le 1, \ \boldsymbol{\Theta}^v = diag\,(\boldsymbol{\theta}) \\ &\quad \boldsymbol{\theta}^T \mathbf{1} = 1, 0 \le \boldsymbol{\theta}_d \le 1, \quad d = 1, \ldots, D_v. \end{aligned} \tag{10}$$

C. Extracting Consistent Information in the Low-Frequency Domain via Low-Rank Tensor

The tensor nuclear norm [46] is effective in extracting consistent information across different modalities; however, high-frequency noise and detrimental information present in hyperspectral data can obscure or even damage the low-rank structure essential for accurate clustering. To address this, we propose an LFTNN (see Definition 6) designed to automatically filter out high-frequency noise and detrimental information, while extracting a consistent low-rank structure.

Definition 6 (LFTNN): Given a tensor $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, where $h = \min(n_1, n_2)$, the LFTNN of $\boldsymbol{\mathcal{A}}$ is defined as

$$\|\boldsymbol{\mathcal{A}}\|_{\text{LFTNN}} = \sum_{i=1}^{n_3} \delta_i \sum_{j=1}^{h} \sigma_j \left(\tilde{\boldsymbol{A}}^i \right) \tag{11}$$

where

$$\delta_i = \begin{cases} \mu, & \lceil n_3 L_f \rceil \le i \le \lceil n_3 \left(1 - L_f\right) \rceil \\ \tau, & \text{otherwise} \end{cases} \tag{12}$$

where $L_f \in (0, 0.5)$, $\mu > \max(\sigma_j(\tilde{\boldsymbol{A}}^i))$, $\tau \ge 0$ denotes regularization parameter, and L_f denotes the low-frequency parameter. The detailed process for computing $\|\boldsymbol{\mathcal{A}}\|_{\text{LFTNN}}$ is presented in Algorithm 1.

This approach combines the low-rank tensor structure with bipartite graph learning. As shown in Fig. 1(e), we stack the bipartite graphs $\boldsymbol{Z}^v$ from different modalities into a tensor $\boldsymbol{\mathcal{Z}} \in \mathbb{R}^{N \times M \times V}$, where the vth frontal slice corresponds to $\boldsymbol{Z}^v$, that is, $\boldsymbol{\mathcal{Z}}(:, :, v) = \boldsymbol{Z}^v$. Before further computations, we first rotate tensor $\boldsymbol{\mathcal{Z}}$, as shown in Fig. 1(e), representing the rotated tensor as $\boldsymbol{\mathcal{Z}}_r \in \mathbb{R}^{N \times V \times M}$. After rotation, $\boldsymbol{\mathcal{Z}}(:, :, v)$ corresponds to $\boldsymbol{\mathcal{Z}}_r(:, v, :) = \boldsymbol{Z}^v$.

Each frontal slice of the original tensor $\boldsymbol{\mathcal{Z}}$ encodes the relationship between all pixels in a single modality and M anchors. For the vth frontal slice of the rotated tensor, $\boldsymbol{Z}_r^v \in \mathbb{R}^{N \times V}$, the ith column specifically represents the relationship between every pixel in the ith modality, and the vth anchor

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Algorithm 1 LFTNN

Input: $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $h = \min(n_1, n_2)$, low frequency parameter L_f, $\tau \geq 0$.
Output: $\|\mathcal{A}\|_{\text{LFTNN}}$.
  1. Compute $\tilde{\mathcal{A}} = fft(\mathcal{A}, [], 3)$.
  2. Compute each frontal slice norm from $\tilde{\mathcal{A}}$ by
  for $i = 1, \ldots, n_3$ **do**
    Compute the singular values $\{\sigma_j(\tilde{\mathbf{A}}^i)\}_{j=1}^{h}$.
    Ensure $\mu > \max(\sigma_j(\tilde{\mathbf{A}}^i))$.
    Sum the singular values: $S_i = \sum_{j=1}^{h} \sigma_j(\tilde{\mathbf{A}}^i)$.
    if $\lceil n_3 L_f \rceil \leq i \leq \lceil n_3(1 - L_f) \rceil$
      Compute frontal slice norm: $\text{SliceNorm}_i = \mu \cdot S_i$.
    else
      Compute frontal slice norm: $\text{SliceNorm}_i = \tau \cdot S_i$.
    end if
  end for
  3. Compute $\|\mathcal{A}\|_{\text{LFTNN}} = \sum_{i=1}^{n_3} \text{SliceNorm}_i$

point associated with that slice. In other words, $\mathbf{Z}_r^v$ already consolidates, at an individual slice level, the connections between the vth anchor point and all pixels across multiple modalities. By considering all such frontal slices together to form the rotated tensor $\mathcal{Z}_r$, we achieve a higher-level, cross-modal integration of the anchor-to-pixel relationships. By minimizing the LFTNN and leveraging tensor singular value decomposition (t-SVD) and FFT, we obtain the common low-rank information among different modalities in the low-frequency domain, as shown in Fig. 1(f). Additionally, since (t-SVD; see Definition 5) operates on each frontal slice individually, computational complexity is intuitively reduced from $\mathcal{O}(NM^2V)$ to $\mathcal{O}(NMV^2)$ postrotation, where $V \ll M$. Therefore, we obtain

$$\begin{aligned}\min_{\mathbf{\Theta}^v, \mathcal{Z}_r} &\sum_{v=1}^{V}\sum_{i=1}^{N}\sum_{j=1}^{M} \left\|\mathbf{\Theta}^v x_i^v - \mathbf{\Theta}^v a_j^v\right\|_2^2 \mathbf{Z}_{ij}^v + \|\mathcal{Z}_r\|_{\text{LFTNN}} \\ \text{s.t. } &\mathbf{Z}^v\mathbf{1} = \mathbf{1}, 0 \leq \mathbf{Z}_{ij}^v \leq 1, \ \mathbf{\Theta} = diag(\boldsymbol{\theta}) \\ &\boldsymbol{\theta}^T\mathbf{1} = 1, 0 \leq \boldsymbol{\theta}_d \leq 1, \quad d = 1, \ldots, D_v. \end{aligned} \tag{13}$$

In summary, the proposed SSBC-LFT method effectively leverages spatial information to construct bipartite graphs, significantly reducing time complexity. Additionally, it efficiently processes multimodal data through a spectral band selection mechanism while simultaneously performing clustering and spectral selection, thereby alleviating the interference from redundant spectral bands. By stacking multimodal data into a tensor and minimizing the LFTNN, our approach explores consistent low-rank structures within low-frequency information while filtering out high-frequency noise and detrimental information, thus enhancing clustering performance.

D. Optimization

Specifically, we employ the augmented Lagrange multiplier (ALM) method [48], which significantly facilitates the optimization process by enabling iterative updates that efficiently converge to the optimal solution. To achieve separability of $\mathcal{Z}_r$, we implement a variable splitting technique, introducing an auxiliary tensor variable $\mathcal{J} \in \mathbb{R}^{N \times V \times M}$ to substitute for $\mathcal{Z}_r$. Consequently, the proposed model can be formulated as

$$\begin{aligned}\min_{\mathbf{\Theta}^v, \mathcal{J}, \mathcal{Z}_r} &\sum_{v=1}^{V}\sum_{i=1}^{N}\sum_{j=1}^{M} \left\|\mathbf{\Theta}^v x_i^v - \mathbf{\Theta}^v a_j^v\right\|_2^2 \mathbf{Z}_{ij}^v \\ &+ \frac{\rho}{2}\left\|\mathcal{J} - \mathcal{Z}_r + \frac{\mathcal{Q}}{\rho}\right\|_F^2 + \|\mathcal{J}\|_{\text{LFTNN}} \\ \text{s.t. } &\mathbf{Z}^v\mathbf{1} = \mathbf{1}, 0 \leq \mathbf{Z}_{ij}^v \leq 1, \ \mathbf{\Theta}^v = diag(\boldsymbol{\theta}) \\ &\boldsymbol{\theta}^T\mathbf{1} = 1, 0 \leq \boldsymbol{\theta}_d \leq 1, \quad d = 1, \ldots, D_v \end{aligned} \tag{14}$$

where $\mathcal{Q} \in \mathbb{R}^{N \times V \times M}$ represents the Lagrange multipliers and $\rho > 0$ denotes the penalty factor.

Notably, (14) involves the joint optimization of three highly coupled variables: $\mathbf{Z}^v$, $\mathbf{\Theta}^v$, and $\mathcal{J}$. Due to the complex interdependencies among these variables, directly solving the joint optimization problem not only incurs high computational cost but also risks convergence to poor local minima. To address this, we adopt an alternating minimization strategy [49], which effectively decomposes the original joint optimization problem (14) into three relatively independent subproblems. These subproblems are then optimized iteratively, allowing for efficient variable optimization.

1) Solving $\mathbf{\Theta}^v$ With Fixed $\mathbf{Z}^v$, $\mathcal{Q}$, and $\mathcal{J}$: In this instance, the optimization of $\mathbf{\Theta}^v$ as presented in (14) evolves into

$$\begin{aligned}\min_{\mathbf{\Theta}^v} &\sum_{v=1}^{V}\sum_{i=1}^{N}\sum_{j=1}^{M} \left\|\mathbf{\Theta}^v x_i^v - \mathbf{\Theta}^v a_j^v\right\|_2^2 \mathbf{Z}_{ij}^v \\ \text{s.t. } &\mathbf{\Theta}^v = diag(\boldsymbol{\theta}), \ \boldsymbol{\theta}^T\mathbf{1} = 1, \quad 0 \leq \boldsymbol{\theta}_d \leq 1, \ d = 1, \ldots, D_v. \end{aligned} \tag{15}$$

By simple matrix algebra, we have

$$\begin{aligned}\min_{\boldsymbol{\theta}_d} &\sum_{v=1}^{V}\sum_{i,j=1}^{N,M}\sum_{d=1}^{D_v} \boldsymbol{\theta}_d^2 \left(x_{id}^v - a_{jd}^v\right)^2 \mathbf{Z}_{ij}^v \\ \text{s.t. } &\mathbf{\Theta}^v = diag(\boldsymbol{\theta}), \boldsymbol{\theta}^T\mathbf{1} = 1, \quad 0 \leq \boldsymbol{\theta}_d \leq 1, \ d = 1, \ldots, D_v. \end{aligned} \tag{16}$$

We define $\mathbf{m}_d = \sum_{v=1}^{V}\sum_{i,j=1}^{N,M}(x_{id}^v - a_{jd}^v)^2\mathbf{Z}_{ij}^v$, (16) can be reformulated as

$$\min_{\boldsymbol{\theta}_d} \sum_{d=1}^{D_v} \boldsymbol{\theta}_d^2 \mathbf{m}_d, \ \text{s.t.} \sum_{d=1}^{D_v} \boldsymbol{\theta}_d = 1, \quad 0 \leq \boldsymbol{\theta}_d \leq 1, \quad d = 1, \ldots, D_v. \tag{17}$$

To solve (17) using the Lagrange multiplier method, we construct the following Lagrangian equation:

$$\mathcal{L} = \sum_{d=1}^{D_v} \boldsymbol{\theta}_d^2 \mathbf{m}_d + \lambda \left(\sum_{d=1}^{D_v} \boldsymbol{\theta}_d - 1\right) \tag{18}$$

where λ represents the Lagrange multiplier. We take the partial derivative of the Lagrangian function $\mathcal{L}$ with respect to the variable $\boldsymbol{\theta}_d$, yielding

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}_d} = 2\boldsymbol{\theta}_d \mathbf{m}_d + \lambda. \tag{19}$$

By setting (19) to zero, we obtain $\boldsymbol{\theta}_d = -\lambda/2\mathbf{m}_d$. Given that $\sum_{d=1}^{D_v} \boldsymbol{\theta}_d = 1$, it follows that $-\lambda/2\sum_{d=1}^{D_v} 1/\mathbf{m}_d = 1$. Simplifying this equation results in $\lambda = -2/\sum_{d=1}^{D_v} 1/\mathbf{m}_d$. Substituting back

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into the expression for $\boldsymbol{\theta}_d$, we obtain the optimal solution for the variable $\boldsymbol{\theta}_d$ as

$$\boldsymbol{\theta}_d^* = \frac{1}{\mathbf{m}_d \sum_{d=1}^{D_v} \frac{1}{\mathbf{m}_d}}. \tag{20}$$

2) Solving $\mathcal{J}$ With Fixed $\mathbf{Z}^v$, $\mathcal{Q}$, and $\boldsymbol{\Theta}^v$: Keeping all other variables fixed, $\mathcal{J}$ updates by addressing the following issue:

$$\min_{\mathcal{J}} \frac{\rho}{2} \left\| \mathcal{J} - \left(\mathcal{Z}_r - \frac{\mathcal{Q}}{\rho} \right) \right\|_F^2 + \|\mathcal{J}\|_{\text{LFTNN}}. \tag{21}$$

To solve the aforementioned equation, which incorporates the proposed LFTNN as a regularization term. We employ the subsequent theorem.

Theorem 1: Consider a tensor $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, where $l = \min(n_1, n_2)$. The tensor $\mathcal{Y}$ can be expressed in its t-SVD form as $\mathcal{Y} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$. In this decomposition, $\mathcal{U}$ and $\mathcal{V}$ are orthogonal tensors that capture the spatial and spectral information, respectively, while $\mathcal{S}$ is a diagonal tensor containing the singular values that represent the significance of the corresponding modes. This representation is crucial for formulating the subsequent optimization problem, which aims to efficiently reconstruct the original tensor $\mathcal{Y}$ while preserving its essential features

$$\min_{\mathcal{X}} \frac{1}{2} \|\mathcal{X} - \mathcal{Y}\|_F^2 + \tau \|\mathcal{X}\|_{\text{LFTNN}}. \tag{22}$$

The optimal solution for (22) can be defined as

$$\mathcal{X}^* = \Gamma_\tau(\mathcal{Y}) = \mathcal{U} * ifft\left(P_\tau\left(\tilde{\mathcal{Y}}\right)\right) * \mathcal{V}^T \tag{23}$$

where $\tilde{\mathcal{Y}} = fft(\mathcal{Y}, [], 3)$ denotes the tensor $\mathcal{Y}$ transformed into the frequency domain through a fast Fourier transform (FFT) along its third mode. The operator $P_\tau(\tilde{\mathcal{Y}})$ is defined as a tensor that facilitates the extraction of specific frequency information, while $P_\tau(\tilde{\mathbf{Y}}^k)$ represents the kth frontal slice of this tensor. This slice is expressed as $P_\tau(\tilde{\mathbf{Y}}^k) = diag(\xi_1, \xi_2, \ldots, \xi_l)$, where each diagonal element ξ_i is calculated using $\xi_i = sign(\sigma_i(\tilde{\mathbf{Y}}^k))\delta_i$. $\sigma_i(\tilde{\mathbf{Y}}^k)$ denotes the ith singular value of the tensor slice $\tilde{\mathbf{Y}}^k$, while δ_i is defined as

$$\delta_i = \begin{cases} max\left(\sigma_i\left(\tilde{\mathbf{Y}}^k\right) - \mu, 0\right), & \lceil n_3 L_f \rceil \le i \le \lceil n_3 \left(1 - L_f\right) \rceil \\ max\left(\sigma_i\left(\tilde{\mathbf{Y}}^k\right) - \tau, 0\right), & \text{otherwise} \end{cases} \tag{24}$$

where $\mu > \max(\sigma_j(\tilde{\mathbf{Y}}^i))$, $\tau \ge 0$ denotes regularization parameter, and L_f denotes the low-frequency parameter.

In the frequency domain, we implement soft thresholding on the singular values of each frontal slice of the tensor $\tilde{\mathbf{Y}}$, denoted as $\tilde{\mathbf{Y}}^k$. The threshold for the high-frequency components, μ, is strategically set to exceed $\max(\sigma_j(\tilde{\mathbf{Y}}^i))$. This ensures that the singular values corresponding to high-frequency slices are effectively reduced to zero, thereby attenuating or eliminating noisy and unstable spectral components. After performing the IFFT to return to the spatial domain, only the structurally meaningful low-frequency information is preserved. This process helps suppress high-frequency noise while enhancing the stability and discriminability of the retained spectral–spatial features.

Based on Theorem 1, the optimal solution to (21) is presented as follows:

$$\mathcal{J}^* = \Gamma_{\frac{1}{\rho}} \left(\mathcal{Z}_r - \frac{\mathcal{Q}}{\rho} \right). \tag{25}$$

Solving $\mathbf{Z}^v$ With Fixed $\mathcal{Q}$, $\mathcal{J}$, and $\boldsymbol{\Theta}^v$. The optimization with respect to $\mathbf{Z}^v$ as delineated in (14) becomes

$$\begin{aligned} \min_{\mathbf{Z}^v} & \sum_{v=1}^{V} \sum_{i=1}^{N} \sum_{j=1}^{M} \left\| \boldsymbol{\Theta}^v x_i^v - \boldsymbol{\Theta}^v a_j^v \right\|_2^2 \mathbf{Z}_{ij}^v \\ & + \frac{\rho}{2} \left\| \mathbf{Z}^v - \mathbf{J}^v - \frac{\mathbf{Q}^v}{\rho} \right\|_F^2 \\ \text{s.t. } & \mathbf{Z}^v \mathbf{1} = \mathbf{1}, \ 0 \le \mathbf{Z}_{ij}^v \le 1. \end{aligned} \tag{26}$$

We define $\mathbf{M}^v = \|\boldsymbol{\Theta}^v x_i^v - \boldsymbol{\Theta}^v a_j^v\|_2^2$ and $\mathbf{P}^v = \mathbf{J}^v + \mathbf{Q}^v/\rho$; thus, (26) can be reformulated as

$$\begin{aligned} \min_{\mathbf{Z}^v} & \sum_{v=1}^{V} \sum_{i=1}^{N} \sum_{j=1}^{M} \mathbf{M}_{ij}^v \mathbf{Z}_{ij}^v + \frac{\rho}{2} \|\mathbf{Z}^v - \mathbf{P}^v\|_F^2 \\ \text{s.t. } & \mathbf{Z}^v \mathbf{1} = \mathbf{1}, 0 \le \mathbf{Z}_{ij}^v \le 1. \end{aligned} \tag{27}$$

By simple matrix algebra, we have

$$\begin{aligned} \min_{\mathbf{Z}^v} & \sum_{v=1}^{V} \langle \mathbf{M}^v - \rho \mathbf{P}^v, \mathbf{Z}^v \rangle + \frac{\rho}{2} \|\mathbf{Z}^v\|_F^2 \\ \text{s.t. } & \mathbf{Z}^v \mathbf{1} = \mathbf{1}, 0 \le \mathbf{Z}_{ij}^v \le 1. \end{aligned} \tag{28}$$

By defining $\mathbf{R}^v = \rho \mathbf{P}^v - \mathbf{M}^v$, we can restate (28) as

$$\begin{aligned} \min_{\mathbf{Z}^v} & \sum_{v=1}^{V} \left\| \mathbf{Z}^v - \frac{1}{\rho} \mathbf{R}^v \right\|_F^2 \\ \text{s.t. } & \mathbf{Z}^v \mathbf{1} = \mathbf{1}, 0 \le \mathbf{Z}_{ij}^v \le 1. \end{aligned} \tag{29}$$

To address this optimization problem, we employ the approach proposed by reference [50]. In (29), each $\mathbf{Z}^v (v = 1, \ldots, V)$ is considered to be independent. For each $\mathbf{Z}^v$, each row $\mathbf{Z}^{v^*}(i,:)$ within the closed-form solution $\mathbf{Z}^{v^*}$ is adjusted to the nearest point on $1/\rho \mathbf{R}^v(i,:)$ within the probability simplex. The probability simplex is defined as the set of all vectors whose elements sum to 1 and are nonnegative.

The entirety of the algorithm is concisely encapsulated in Algorithm 2.

E. Complexity Analysis

In evaluating the time complexity of SSBC-LFT, our analysis primarily centers on the ERS segmentation algorithm and the key computational steps involved in solving problem (13). Specifically, we focus on analyzing the computational complexity of the ERS segmentation algorithm and the update procedures for the variables $\{\boldsymbol{\Theta}^v\}_{v=1}^V$, $\mathcal{J}$, and $\{\mathbf{Z}^v\}_{v=1}^V$. Their respective time complexities are $\mathcal{O}(N \log N)$, $\mathcal{O}\left(NMV \sum_{v=1}^{V} D_v\right)$, $\mathcal{O}(NMV(\log M + V))$, and $\mathcal{O}\left(NM\left(\sum_{v=1}^{V} D_v + V \log M\right)\right)$, where $\boldsymbol{D}_v$ represents the number of bands in each modality and $V \ll \{\boldsymbol{D}_v, M\} \ll N$. Aggregating all subproblem complexities, the time complexity of SSBC-LFT is $\mathcal{O}\left(NM(V+1) \sum_{v=1}^{V} D_v\right)$.

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Algorithm 2 SSBC-LFT

Input: HSI: $\mathcal{X}^1 \in \mathbb{R}^{W\times H\times D_1}$, superpixels number M, LiDAR data: $\mathcal{X}^2 \in \mathbb{R}^{W\times H\times D_2}$, and cluster number c.
Perform ERS to generate HSI & LiDAR superpixels.
Generate anchor matrix $\boldsymbol{A}^v = [\boldsymbol{a}_1^v, \cdots, \boldsymbol{a}_M^v] \in \mathbb{R}^{D_v\times M}$.
Generate initial bipartite graph $\{\boldsymbol{Z}^v\}_{v=1}^V \in \mathbb{R}^{N\times M}$.
Construct multimodal tensor $\mathcal{Z} \in \mathbb{R}^{N\times M\times V}$ and rotate;
Initialize $\mathcal{Q} = \mathcal{J} = \mathbf{0}$, $\rho = 10^{-5}$, max $_\rho = 10^{10}$, $\eta = 1.1$, $\boldsymbol{\Theta}^v = diag\left(\frac{1}{D_v}, \frac{1}{D_v}, \ldots, \frac{1}{D_v}\right)$;
repeat
Calculate $\{\boldsymbol{\Theta}^v\}_{v=1}^V$ by solving Eq. (14);
Calculate $\mathcal{J}$ by solving Eq. (16);
Calculate $\{\boldsymbol{Z}^v\}_{v=1}^V$ by solving Eq. (20);
Calculate $\mathcal{Q}$ and ρ: $\mathcal{Q} = \mathcal{Q} + \rho(\mathcal{Z}_r - \mathcal{J})$, $\rho = \min(\eta \times \rho, \max_\rho)$;
until $\Delta obj < 10^{-5}$ or maximum iteration
Use the K-means algorithm on $\boldsymbol{Z} = \sum_{v=1}^V \frac{\boldsymbol{Z}^v}{V}$ to determine the clustering labels;
Output: The final clustering result.

V. Experiment

To evaluate the effectiveness of the proposed SSBC-LFT method, we conducted experiments on the Trento, MUUFL, and Houston datasets. The results demonstrate that SSBC-LFT significantly outperforms baseline methods across multiple accuracy metrics, including pixel accuracy (PA), average accuracy (AA), overall accuracy (OA), and the Kappa. Additionally, it exhibits superior computational efficiency compared to most baseline approaches. All experiments were conducted in MATLAB 2022a on a Windows 11 PC with an Intel i7-13700 CPU and 32GB RAM.

A. Datasets

The specific characteristics of the three HSI-LiDAR datasets utilized in this study are outlined below.

1) *Trento [51]:* Collected over a rural area south of Trento, Italy, this dataset comprises co-registered HSI and LiDAR data with a spatial size of 600 × 166 pixels and a 1-m resolution. The HSI, acquired by the AISA Eagle sensor, contains 63 bands spanning 0.40–0.98 μm, while the LiDAR data, captured by the Optech ALTM 3100EA sensor, provides two channels. The dataset includes six land-cover classes.
2) *MUUFL*[1]: Acquired at the University of Southern Mississippi Gulfport Campus (Long Beach, MS, USA), this dataset contains co-registered HSI and LiDAR data. The HSI, captured by the CASI-1500 sensor, has 325 × 220 pixels with 64 effective bands spanning 375–1050 nm at 10-nm intervals and a spatial resolution of 0.54 × 1.0 m. The LiDAR-derived DSM, collected by the Gemini ALTM sensor, offers a spatial resolution of 0.60 × 0.78 m. The scene comprises 11 land-cover classes.

[1]https://github.com/GatorSense/MUUFLGulfport

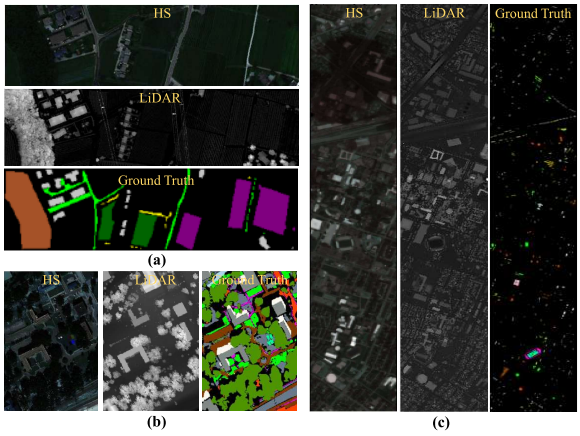


Fig. 2. False-color visualizations of HSI modality, LiDAR modality, and ground truth for three datasets. Black denotes unknown classes for each ground-truth image. (a) Trento. (b) MUUFL. (c) Houston.

3) *Houston*[2]*:* Acquired at the University of Houston and surrounding areas, this dataset includes co-registered HSI and LiDAR-derived DSM data, each with 349 × 1905 pixels and a spatial resolution of 2.5 m. The HSI comprises 144 bands spanning 0.38–1.05 μm, while the LiDAR data include a single channel, covering 15 land-cover classes.

The details of the dataset information and land-cover types, along with the visualization of these datasets, are provided in Tables I and II, and illustrated in Fig. 2, respectively.

B. Comparison With Other Algorithms

1) *RCAGL (Liu et al. [52]):* RCAGL constructs a consistent anchor graph by jointly learning a shared and a view-specific component, capturing intermodal consistency while filtering out view-specific noise.
2) *CAMVC (Zhang et al. [53]):* CAMVC employs an adaptive cluster-wise anchor learning strategy for multimodal RS clustering, enforcing cluster structures on latent anchors via a consensus indicator matrix.
3) *AWMVC (Wan et al. [54]):* AWMVC leverages matrix factorization to learn coefficient matrices from multimodal RS data, integrating them into a unified consensus matrix for optimal results.
4) *FastMICE (Huang et al. [55]):* FastMICE employs random modal groups to model cross-modal relationships, enabling a hybrid early–late fusion strategy for efficient multistage multimodal integration.
5) *FDAGF (Zhang et al. [56]):* FDAGF improves clustering by jointly optimizing the weights of multimodal multi-choice anchor graphs and the significance of selections within each modality.
6) *FPMVS_CAG (Wang et al. [57]):* FPMVS_CAG integrates anchor learning and graph construction into a co-optimized framework, enhancing multimodal fusion through a unified consensus anchor graph.

[2]http://www.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

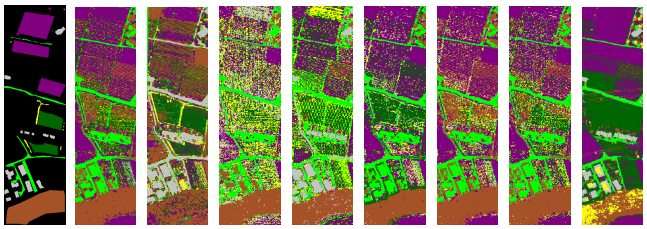
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TABLE I
INTRODUCTION OF THE DATASETS

Dataset	Sensor Type	Dimensions (pixels)	Spectral Channels	Spectral Range (μm)	Spatial Resolution (m)	Categories
Trento	HSI&LiDAR	600 x 166	63	0.4000 – 0.9800	1	6
MUUFL	HSI&LiDAR	325 x 220	64	0.3750 – 1.0500	0.5400 x 1.000 (HSI), 0.6000 x 0.7800 (LiDAR)	11
Houston	HSI&LiDAR	349 x 1905	144	0.3800 – 1.0500	2.5000	15

TABLE II
LAND-COVER TYPES OF THE TRENTO, MUUFL, AND HOUSTON

Trento		MUUFL		Houston	
Class Name	#Samples	Class Name	#Samples	Class Name	#Samples
Apple trees	4034	Trees	23246	Healthy grass	1251
Buildings	2903	Mostly grass	4270	Stressed grass	1254
Ground	479	Mixed ground surface	6882	Synthetic grass	697
Wood	9123	Dirt and sand	1826	Trees	1244
Vineyard	10501	Road	6687	Soil	1242
Roads	3174	Water	466	Water	325
		Building shadow	2233	Residential	1268
		Building	6240	Commercial	1244
		Sidewalk	1385	Road	1252
		Yellow curb	183	Highway	1227
		Cloth Panels	269	Railway	1235
				Parking Lot 1	1233
				Parking Lot 2	469
				Tennis Court	428
				Running Track	660
Total	30214	Total	53687	Total	15029



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 3. Clustering performance on the Trento dataset. (a) Ground-truth result. (b)–(i) Results of algorithms RCAGL, CAMVC, AWMVC, FastMICE, FDAG, FPMVS_CAG, SMVSC, and SSBC-LFT, respectively.

7) *SMVSC (Sun et al. [58]):* SMVSC unifies orthogonal consensus anchor learning and anchor graph construction, producing an optimal consensus anchor graph with improved class structure clarity.

C. Experimental Results on the Trento Dataset

In this section, we present an extensive evaluation of our proposed SSBC-LFT framework on the Trento dataset, which is characterized by high-dimensional hyperspectral data with spatially clustered classes such as vineyards, apple orchards, and woodlands. Our experiments demonstrate that the incorporation of spatial dependencies, adaptive spectral band selection, and LFTNN regularization not only enhances clustering accuracy but also substantially improves noise robustness and computational efficiency. Table III reports the performance in terms of producer's accuracy (PA), AA, OA, and the Kappa coefficient (Kappa), alongside the CPU time. Fig. 3 illustrates the visualization of clustering results obtained by SSBC-LFT and baseline algorithms on the Trento dataset.

The primary innovation of our method lies in its *spatially enriched bipartite graph construction*. By integrating spatial coherence into the clustering process, our framework preserves local spatial continuity, which is crucial for HSIs where objects such as vineyards and woodlands form contiguous regions. Unlike traditional techniques that treat pixels independently, our approach dynamically adjusts neighborhood relationships through graph-based modeling, resulting in more homogeneous clusters. The results in Table III indicate that our method outperforms both RCAGL and CAMVC, as well as conventional methods such as AWMVC and FastMICE, achieving an OA of 0.7842, an AA of 0.6626, and a Kappa of 0.7241. These improvements underscore the strength of spatially informed clustering, which reduces fragmented regions and misclassifications.

Another critical component is the *adaptive spectral band selection*. HSIs contain many spectral bands, some of which contribute redundant or noisy information. Our algorithm assigns higher weights to the most informative bands while suppressing the influence of less relevant ones. This strategy is particularly effective in complex clusters such as wood and buildings. By filtering out noise and minimizing spectral redundancy, our framework ensures robust clustering.

Furthermore, our LFTNN regularization plays a vital role in suppressing high-frequency noise, stabilizing the clustering process, and enhancing robustness. This is particularly beneficial in the roads class, where reducing the influence of high-frequency components helps delineate subtle boundaries in noisy environments.

From a computational perspective, our framework balances accuracy and efficiency. Despite incorporating advanced graph construction and spectral band selection mechanisms, our algorithm achieves a runtime of 80.9531 s, outperforming several baseline models. This efficiency is attributed to the reduced number of anchor points derived from ERS-based segmentation, streamlining the clustering process without compromising spatial detail.

In summary, the experimental results on the Trento dataset confirm that our SSBC-LFT method significantly outperforms existing approaches by effectively exploiting spatial information, selectively harnessing spectral data, and mitigating noise through low-frequency regularization. The integration of RCAGL and CAMVC into our comparative analysis further validates the robustness and adaptability of our framework in handling high-dimensional hyperspectral data. Consistent improvements across various metrics, as shown in Table III, highlight the potential of our method for real-world RS applications where both spatial coherence and spectral precision are essential.

D. Experimental Results on the MUUFL Dataset

The MUUFL dataset presents a challenging hyperspectral clustering task due to its diverse land cover types, including

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TABLE III
QUANTITATIVE EVALUATIONS ON THE TRENTO DATASET. THE BEST VALUE IN EACH ROW IS IN BOLD (TIME IS MEASURED IN SECONDS)

Class	RCAGL	CAMVC	AWMVC	FastMICE	FDAGF	FPMVS_CAG	SMVSC	SSBC-LFT
Apple trees	0.5884	0.4694	0.4114	0.4060	0.4192	0.5918	0.5618	**0.6514**
Buildings	0.6667	0.5011	0.0006	0.4211	0.0000	0.6754	0.0000	**0.9241**
Ground	0.0111	**0.5714**	0.0268	0.0391	0.0235	0.0267	0.0340	0.0000
Vineyard	0.7662	0.5733	0.8650	**0.9668**	0.9483	0.7603	0.8394	0.9515
Wood	0.9060	0.6285	**0.9775**	0.9200	0.9027	0.9037	0.9405	0.9720
Roads	0.5102	0.5382	0.6589	0.6232	0.4952	0.5150	0.5154	**0.9922**
AA	0.5266	0.6150	0.4896	0.5824	0.5164	0.5318	0.5380	**0.6626**
OA	0.7100	0.5470	0.5793	0.6101	0.7026	0.7093	0.6984	**0.7842**
Kappa	0.6147	0.4149	0.4779	0.5196	0.6115	0.6134	0.6071	**0.7241**
Cpu Time	116.3293	15.7522	**15.5676**	138.2671	400.9348	517.1013	258.5940	80.9531



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 4. Clustering performance on the MUUFL dataset. (a) Ground-truth result. (b)–(i) Results of algorithms RCAGL, CAMVC, AWMVC, FastMICE, FDAG, FPMVS_CAG, LMVSC, and SSBC-LFT, respectively.

TABLE IV
QUANTITATIVE EVALUATIONS ON THE MUUFL DATASET. THE BEST VALUE IN EACH ROW IS IN BOLD (TIME IS MEASURED IN SECONDS)

Class	RCAGL	CAMVC	AWMVC	FastMICE	FDAGF	FPMVS_CAG	SMVSC	SSBC-LFT
Trees	0.9237	0.9770	0.9561	0.9878	0.9787	0.6287	0.6876	**0.9879**
Mostly grass	0.4612	0.4611	**0.5326**	0.1285	0.5279	0.2176	0.3384	0.4832
Mixed ground surface	0.3519	0.6528	0.6392	0.5668	0.5371	**0.8162**	0.7095	0.5845
Dirt and sand	0.0769	0.2155	0.1855	0.0878	0.2985	0.0072	0.0427	**0.6847**
Road	0.5792	0.7068	0.4992	0.8661	0.3485	0.3485	0.6851	**0.8889**
Water	0.0000	**0.2857**	0.0305	0.1376	0.0082	0.0013	0.0007	0.0000
Building shadow	0.1301	**0.5650**	0.1919	0.1966	0.4835	0.0098	0.1361	0.1962
Building	0.8414	0.7537	0.2636	0.6408	0.9098	0.9096	0.9145	**0.9163**
Sidewalk	0.0000	0.0000	0.0562	0.0037	0.0357	0.0103	0.0092	**0.2444**
Yellow curb	0.0000	0.0224	0.0005	0.0000	0.0046	**0.0635**	0.0137	0.0142
Cloth Panels	0.0000	0.0534	0.0000	0.0619	0.0227	0.0000	0.0000	**0.9632**
AA	0.3180	0.4858	0.3035	0.3935	0.4412	0.2259	0.2445	**0.5098**
OA	0.4788	0.6138	0.3500	0.3415	0.4760	0.3998	0.3897	**0.6172**
Kappa	0.3733	0.5273	0.2705	0.2550	0.4038	0.2479	0.2529	**0.5390**
Cpu Time	68.6403	21.1564	**14.3504**	16.3626	187.7522	109.8084	68.9494	42.5057

vegetation, urban structures, and mixed ground surfaces. Our method demonstrates significant improvements in clustering accuracy and computational efficiency compared to existing approaches, as shown in Table IV. Fig. 4 shows the clustering results of SSBC-LFT and baselines on the MUUFL dataset.

One key innovation of our method is the *spatially enriched bipartite graph construction*, which effectively integrates spatial and spectral information. This strategy enables better segmentation of classes with complex structures, such as trees and mixed ground surfaces, where our model attains high PA scores. By enforcing spatial coherence, the method reduces misclassification at boundaries, a common issue in hyperspectral clustering.

Another crucial component is the *adaptive spectral band selection*, which prioritizes discriminative spectral bands while discarding redundant ones. This feature selection enhances clustering performance in smaller yet critical categories such as dirt and sand, where conventional methods struggle due to spectral overlap. Notably, our method achieves superior performance in these minor classes without excessive computational costs.

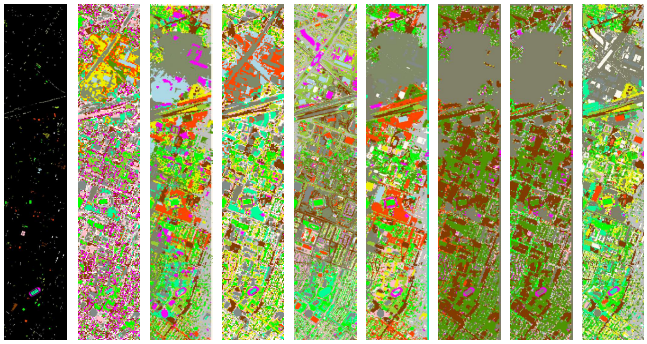
Moreover, the LFTNN significantly improves robustness against high-frequency noise. This enhancement is particularly beneficial for urban categories such as roads and building shadows, where high-frequency spectral variations often degrade clustering performance. Our approach effectively suppresses noise, leading to substantial gains in segmentation accuracy over competing methods like AWMVC and FDAGF.

From a computational standpoint, our model balances accuracy and efficiency. With a runtime of 42.5057 s, it remains competitive while outperforming recent methods such as RCAGL and CAMVC in overall clustering metrics. The

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TABLE V
QUANTITATIVE EVALUATIONS ON THE HOUSTON DATASET. THE BEST VALUE IN EACH ROW IS IN BOLD (TIME IS MEASURED IN SECONDS)

Class	RCAGL	CAMVC	AWMVC	FastMICE	FDAGF	FPMVS_CAG	SMVSC	SSBC-LFT
Healthy grass	0.0000	0.5241	0.5173	0.4421	0.5062	0.0000	0.0000	**0.5590**
Stressed grass	0.0000	0.5173	**0.7362**	0.3876	0.7108	0.0000	0.0000	0.3033
Synthetic grass	0.0000	**0.7318**	0.4306	0.3840	0.6402	0.0000	0.0000	0.5095
Trees	0.2986	0.3158	0.3095	**0.7363**	0.0186	0.3096	0.3176	0.5777
Soil	0.4613	0.7147	0.4393	0.2175	0.5037	0.2754	0.2962	**0.8548**
Water	0.0000	0.0730	**0.1568**	0.0477	0.0309	0.0000	0.1224	0.0000
Residential	0.4595	0.5849	0.4768	0.4320	0.5488	0.6154	**0.6512**	0.3810
Commercial	0.2934	0.7220	0.9136	0.3858	**0.9854**	0.7838	0.1937	0.4149
Road	0.6022	0.5231	0.4678	0.5287	0.4491	0.7616	**0.8038**	0.5145
Highway	0.0714	**0.5348**	0.2903	0.3818	0.2864	0.0000	0.4667	0.4360
Railway	0.1933	0.2942	0.4427	0.3984	0.4091	0.1971	0.1825	**0.4704**
Parking Lot1	0.0000	0.3667	0.3113	0.3011	0.2557	0.3797	0.2111	**0.4062**
Parking Lot2	0.0000	0.1714	0.0192	0.0537	0.0000	0.4531	**0.5385**	0.0000
Tennis Court	0.7111	0.0476	0.0000	0.2380	0.3994	**0.7212**	0.5840	0.3712
Running Track	0.0000	0.6198	0.3825	0.0000	0.7169	**0.7213**	0.6636	0.5762
AA	0.2533	0.4607	0.4232	0.3613	0.4422	0.3000	0.3352	**0.4944**
OA	0.2937	0.4929	0.4048	0.3675	0.4458	0.2947	0.3798	**0.5001**
Kappa	0.2313	0.4531	0.3621	0.3214	0.4012	0.2351	0.3264	**0.4610**
Cpu Time	204.1015	211.7268	381.6968	**209.6567**	4243.3208	4954.4023	2185.6886	468.4316



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 5. Clustering performance on the Houston dataset. (a) Ground-truth result. (b)–(i) Results of algorithms RCAGL, CAMVC, AWMVC, FastMICE, FDAG, FPMVS_CAG, SMVSC, and SSBC-LFT, respectively.

efficiency gains stem from our optimized anchor point selection and adaptive spectral band weighting, which reduce complexity without compromising accuracy.

Overall, the experimental results confirm that our approach consistently surpasses alternative methods in AA, OA, and Kappa scores. The combination of spatially aware graph modeling, adaptive feature selection, and noise suppression makes our method particularly suitable for real-world hyperspectral clustering applications.

E. Experimental Results on the Houston Dataset

The Houston dataset presents a significant challenge due to its high dimensionality and diverse urban and vegetative classes. Table V summarizes our experimental results, comparing our SSBC-LFT method against recent clustering algorithms, including RCAGL and CAMVC. Fig. 5 displays the clustering results of SSBC-LFT and baselines on the Houston dataset.

Our method's effectiveness is primarily attributed to the *spatially enriched bipartite graph construction*, which integrates spatial continuity into the clustering process. This enhances accuracy in structurally complex classes such as healthy grass, residential, and parking lot1, where background variability is high. As shown in Table V, our method achieves the highest PA-scores in healthy grass (0.5590) and parking lot1 (0.4062), surpassing other algorithms by effectively preserving spatial coherence.

The second key innovation is the *adaptive spectral band selection*, which reduces spectral redundancy and noise, particularly in high-dimensional classes such as highway and commercial. By focusing on the most discriminative spectral bands, our method improves class separability, achieving a PA-score of 0.4360 in highway—higher than most baselines. This mechanism enhances clustering robustness by suppressing irrelevant spectral variations.

Furthermore, our LFTNN effectively mitigates high-frequency noise, which is prevalent in urban areas due to surface heterogeneity. This leads to well-defined clusters in categories such as soil and railway, where our method achieves superior PA-scores (0.8548 and 0.4704, respectively), outperforming competing approaches.

Regarding computational efficiency, our method completes clustering in 468.4316 s. While not the fastest, this runtime remains competitive considering the accuracy gains and robustness improvements. The efficiency stems from optimized anchor point selection and spectral filtering, which reduce computational overhead.

Overall, the experimental results validate the superiority of our SSBC-LFT framework, which consistently outperforms competing methods, including the newly incorporated RCAGL and CAMVC, demonstrating its efficacy for large-scale, complex RS data analysis.

Across the Trento, MUUFL, and Houston datasets, our SSBC-LFT method consistently outperforms other techniques in clustering accuracy, noise robustness, and computational

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TABLE VI

ABLATION STUDY RESULTS ON BAND SELECTION AND LFTNN. THE BEST VALUE IN EACH COLUMN IS IN BOLD

Dataset	BS	LFTNN	AA	OA	Kappa
Trento	×	×	0.5379	0.6587	0.5599
	×	✓	0.6130	0.7105	0.6361
	✓	×	0.5878	0.6907	0.6078
	✓	✓	**0.6626**	**0.7842**	**0.7241**
MUUFL	×	×	0.2018	0.4103	0.2394
	×	✓	0.4546	0.5346	0.4540
	✓	×	0.4533	0.5264	0.4522
	✓	✓	**0.5098**	**0.6172**	**0.5390**
Houston	×	×	0.4888	0.4937	0.4512
	×	✓	0.4867	0.5197	0.5197
	✓	×	0.4922	0.4924	0.4542
	✓	✓	**0.4944**	**0.5001**	**0.4610**

efficiency. This success stems from three key innovations: first, the *spatially enriched bipartite graph construction*, which incorporates spatial dependencies to improve cluster cohesion and boundary accuracy, especially in contiguous clusters like vegetation or urban areas. Second, the *adaptive spectral band selection*, which dynamically chooses the most informative bands, enhancing accuracy while reducing computational cost, particularly in high-dimensional datasets. Third, the LFTNN, which filters out high-frequency noise and focuses on stable low-rank structures, boosts performance in challenging categories. These innovations collectively enable superior clustering results, outperforming state-of-the-art methods in unsupervised multimodal RS applications, as demonstrated by the quantitative results in Tables III–V.

F. Ablation Studies

In this section, we conduct a comprehensive analysis to evaluate the effectiveness of our proposed clustering method. Specifically, we examine the contributions of our three key innovations: spatially enriched bipartite graph construction, adaptive spectral band selection, and the LFTNN. Additionally, we compare the performance of different superpixel segmentation methods, including two newly introduced algorithms: simple noniterative clustering (SNIC) and Watershed segmentation. The experiments are performed on three benchmark datasets, and the results substantiate the superiority of our approach.

1) Effectiveness of Adaptive Spectral Band Selection: The impact of our adaptive spectral band selection strategy is detailed in Table VI. This method dynamically assigns higher weights to informative spectral bands while suppressing less relevant ones, thereby improving clustering accuracy. The experimental results indicate that the inclusion of band selection consistently enhances classification performance. For instance, on the MUUFL dataset, the adaptive band selection strategy yields a notable improvement (AA: **0.5098**, OA: **0.6172**, and Kappa: **0.5390**). By selectively utilizing the most discriminative spectral bands, our approach mitigates the curse of dimensionality, leading to more robust and accurate clustering.

2) Evaluation of the LFTNN: Table VI further illustrates the benefits of the LFTNN. By effectively filtering out high-frequency noise and preserving low-rank structures, the LFTNN contributes to significant performance gains. On the Houston dataset, the model incorporating the LFTNN achieves the highest clustering accuracy (AA: **0.4944**, OA: **0.5001**, and Kappa: **0.4610**). This demonstrates that suppressing high-frequency variations while emphasizing stable low-rank structures is essential for improving clustering robustness, particularly in heterogeneous hyperspectral datasets.

TABLE VII

ABLATION STUDY RESULTS ON SEGMENTATION ALGORITHMS. THE BEST VALUE IN EACH COLUMN IS IN BOLD (TIME IS MEASURED IN SECONDS)

Dataset	Segmentation	AA	OA	Kappa	Segment Time
Trento	SLIC	0.5764	0.6713	0.5870	0.1882
	LSC	0.6081	0.7063	0.6293	0.2296
	SNIC	0.5891	0.6823	0.6001	**0.1865**
	Watershed	0.5969	0.7012	0.6208	0.2034
	ERS	**0.6626**	**0.7842**	**0.7241**	0.3517
MUUFL	SLIC	0.4370	0.5217	0.4473	0.0947
	LSC	0.4743	0.5081	0.4324	0.1420
	SNIC	0.4335	0.5187	0.4443	**0.0895**
	Watershed	0.3840	0.5577	0.4505	0.1154
	ERS	**0.5098**	**0.6172**	**0.5390**	0.1776
Houston	SLIC	0.4080	0.4654	0.4218	2.9206
	LSC	0.4537	0.4663	0.4257	**2.8340**
	SNIC	0.4490	0.4620	0.4490	3.7151
	Watershed	0.4343	0.4698	0.4283	3.1140
	ERS	**0.4944**	**0.5001**	**0.4610**	4.7077

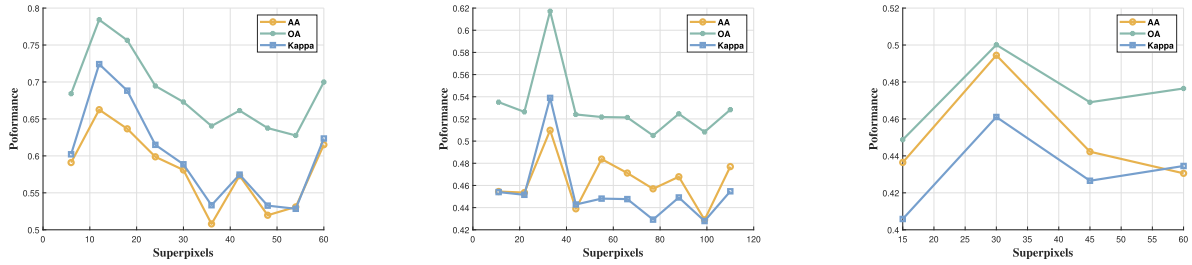
3) Comparison of Superpixel Segmentation Methods: The comprehensive comparative analysis of various state-of-the-art superpixel segmentation methods, including SLIC [59], LSC [60], SNIC [61], Watershed [62], and the ERS algorithm, is systematically presented in Table VII. The experimental results clearly indicate that ERS consistently outperforms alternative segmentation techniques across all evaluated datasets, achieving the highest segmentation accuracy and overall classification performance (e.g., Trento dataset: AA: **0.6626**, OA: **0.7842**, and Kappa: **0.7241**). Notably, the recently introduced SNIC and Watershed algorithms exhibit competitive segmentation performance while maintaining significantly lower computational complexity. However, ERS remains the superior approach due to its enhanced capability to effectively capture intricate spatial relationships within hyperspectral imagery, which is crucial for precise segmentation and subsequent classification tasks. These empirical findings underscore the pivotal role of superpixel segmentation strategies in improving clustering performance, thereby reinforcing the overall effectiveness and robustness of our proposed methodological framework.

Furthermore, the results of the ablation study provide substantial validation for the efficacy of our proposed innovations, demonstrating their significant contribution to enhanced clustering accuracy, robustness, and generalization across diverse hyperspectral and multimodal RS datasets.

G. Number of Superpixels

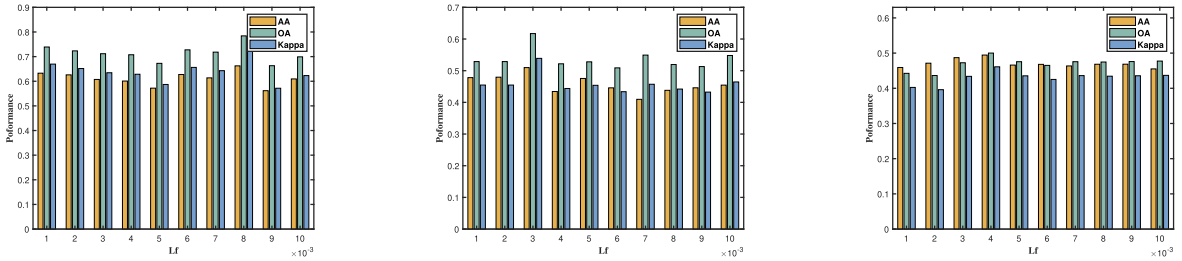
The variations in model performance relative to the quantity of superpixels are depicted in Fig. 6. These results reveal that

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(a) (b) (c)

Fig. 6. Performance sensitivity to the number of superpixels for different data types. While performance initially improves with more superpixels, the relationship is not linear, and further increases can cause erratic results, indicating that optimal performance is achieved by effectively selecting representative superpixels without over-segmentation. (a) Trento. (b) MUUFL. (c) Houston.



(a) (b) (c)

Fig. 7. Effect of the L_f parameter on clustering performance across three datasets, showing an initial improvement as low-frequency information is retained, followed by a decline due to high-frequency noise, illustrating the importance of balancing information retention and noise reduction. (a) Trento. (b) MUUFL. (c) Houston.

there is no straightforward linear correlation between superpixel count and model effectiveness; thus, merely increasing superpixel numbers does not equate to improved performance. As observed in Fig. 6, performance indicators—such as AA, OA, and Kappa—do show initial gains as superpixel quantity rises. However, substantial variability is apparent across the spectrum of superpixel values. For instance, Fig. 6(a) illustrates that Kappa reaches its maximum around a superpixel count of 10. Beyond this point, further increases lead to inconsistent results, sometimes even diminishing performance. This indicates that the proposed approach can effectively pinpoint the most representative superpixels without excessive segmentation, thereby preserving both efficiency and accuracy. Importantly, this outcome underscores the model's robustness in optimizing superpixel selection while minimizing over-segmentation, which could otherwise introduce noise and redundancy detrimental to performance.

H. Parameter Sensitivity Analysis

The performance of our clustering model is sensitive to the parameter L_f, which controls the amount of information retained in the low-frequency domain. As shown in Fig. 7, an increase in L_f initially enhances clustering performance metrics—OA, AA, and Kappa—across all three datasets (Trento, MUUFL, and Houston). This trend indicates that when L_f is small, insufficient low-frequency information is retained, limiting the model's ability to capture essential spatial and spectral structures. However, as L_f continues to increase, we observe a turning point where performance declines. This is due to excessive high-frequency information, which adds noise and reduces clustering quality.

Our model's design addresses this sensitivity by effectively balancing information retention and noise reduction. By using an adaptive mechanism for band selection and LFTNN regularization, our method can optimize L_f to retain critical information while minimizing the adverse effects of high-frequency noise. This approach allows the model to achieve robust and reliable clustering performance without the pitfalls of over-segmentation or the inclusion of redundant information.

I. Generalization Experiments

Our proposed method demonstrates strong clustering performance on the HSI-LiDAR multimodal RS dataset, further highlighting its broad adaptability and potential for practical RS applications. To evaluate its effectiveness in more complex scenarios, we selected the Augsburg [63] dataset, which presents higher heterogeneity, for further generalization experiments. This dataset integrates information from three distinct modalities: hyperspectral, synthetic aperture radar (SAR), and digital surface model (DSM), encompassing seven typical land-cover types. We performed a systematic comparison between SSBC-LFT and baseline algorithms, with the results shown in Table VIII. The experiments show that SSBC-LFT achieves the highest OA of **0.6399** and a Kappa of **0.5313**, significantly outperforming the competitors. This superior performance underscores the method's ability to effectively integrate spatial and spectral features for enhanced

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TABLE VIII
QUANTITATIVE EVALUATIONS ON THE AUGSBURG DATASET. THE BEST VALUE IN EACH ROW IS IN BOLD (TIME IS MEASURED IN SECONDS)

Evaluation Metric	RCAGL	CAMVC	AWMVC	FastMICE	FDAGF	FPMVS_CAG	SMVSC	SSBC-LFT
AA	0.2076	**0.4594**	0.3209	0.4263	0.3855	0.2562	0.2610	0.4300
OA	0.4600	0.5605	0.2730	0.5068	0.4142	0.4285	0.3864	**0.6399**
Kappa	0.2148	0.4566	0.1284	0.4084	0.2892	0.2066	0.1667	**0.5313**
CPU Time	258.0341	53.3363	30.8109	**30.5933**	507.8588	168.8679	133.4325	156.4836

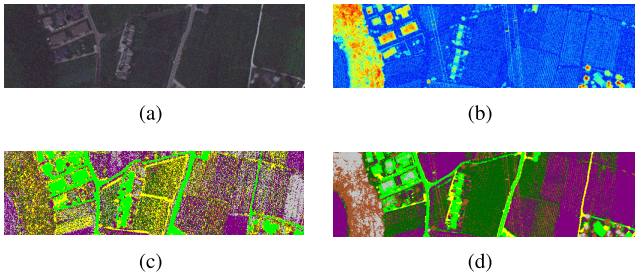


Fig. 8. Visualization of the noise experiment. (a) Noisy HSI. (b) Noisy LiDAR image. (c) Clustering results by CAMVC (Kappa = 0.2943). (d) Clustering results by SSBC-LFT (Kappa = 0.5492).

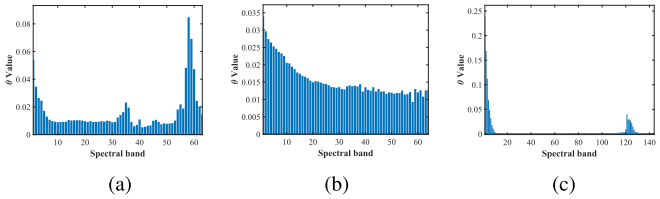


Fig. 9. Visualization experiments on spectral band significance on diverse multimodal RS datasets. (a) Trento. (b) MUUFL. (c) Houston.

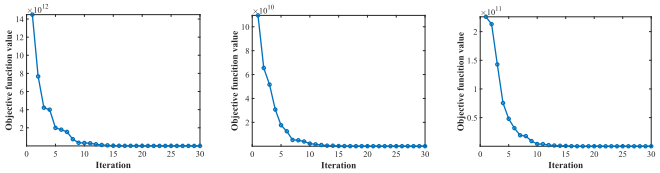


Fig. 10. Convergence experiments on diverse multimodal RS datasets. (a) Trento. (b) MUUFL. (c) Houston.

classification. In terms of computational efficiency, our method completes the clustering task in 156.48 s, balancing accuracy and computational cost. Although not the fastest, it remains considerably more efficient than FDAGF (507.86 s). These results validate our method's robustness and effectiveness, reinforcing its potential for advanced RS data analysis.

J. Noise Robustness Experiments

To assess the noise robustness of our SSBC-LFT model, we introduce Gaussian noise with a mean of 1 and a standard deviation of 0.05 into each modality of the Trento HS-LiDAR dataset. We then compare our method with CAMVC, a state-of-the-art multimodal clustering approach. Fig. 8 presents the noisy input data and clustering results under noise conditions.

As shown in Fig. 8(c), CAMVC struggles with noise, producing fragmented clusters, particularly in the central and lower left regions, where numerous small isolated pixel groups emerge. This suggests that CAMVC lacks sufficient spatial consistency in noisy settings. In contrast, Fig. 8(d) shows that SSBC-LFT maintains well-structured clusters, effectively preserving spatial continuity. This advantage arises from our spatially informed bipartite graph, which enhances local consistency by grouping proximate pixels into the same cluster.

Furthermore, the CAMVC result exhibits large-scale misclassification with irregular color distributions, leading to instability. In contrast, SSBC-LFT demonstrates the high consistency across classified regions, significantly reducing noise effects. Quantitatively, our method achieves a Kappa of 0.5492, surpassing CAMVC's 0.2943. This improvement stems from our LFTNN regularization, which extracts cross-modal low-rank consistency.

In summary, the results confirm that SSBC-LFT exhibits superior noise robustness, achieving more stable and accurate clustering than CAMVC. This highlights the effectiveness of our spatially informed bipartite graph and LFTNN-based noise suppression in real-world RS applications where robustness is critical.

K. Spectral Band Significance Analysis

In the spectral band significance visualization experiment, we visualized the spectral band weight matrix $\boldsymbol{\Theta}$ of the HSI modality learned by the proposed SSBC-LFT model. As shown in Fig. 9, for the Trento dataset [Fig. 10(a)], the model assigns higher weights to the low-band region (bands 1–5) and the high-band region (bands 55–64), while the middle bands receive relatively low weights. For the MUUFL dataset [Fig. 10(b)], the weight distribution exhibits a "high-front–low-back" pattern, with significantly higher weights in the first 20 low bands and lower weights in the middle and high bands. In the Houston dataset [Fig. 10(c)], the learned weight distribution shows a distinct bimodal pattern, with high weights concentrated in the low-band region (bands 1–10) and the high-band region (bands 115–130), while most intermediate bands receive near-zero weights.

This phenomenon is highly consistent with the spectral characteristics of each dataset. Specifically, in the Trento dataset, vegetation types such as vineyards, woodlands, and apple trees exhibit strong reflectance in the high spectral bands corresponding to the near-infrared region, while nonnatural targets are more easily distinguished in the low spectral bands within the visible spectrum. In the MUUFL dataset, the visible spectrum, particularly the blue and green regions, facilitates

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the discrimination of complex land cover classes such as vegetation, man-made structures, and interference-prone areas, thereby enhancing interclass separability. The Houston dataset contains a wide variety of land cover types. The visible bands provide fine discrimination of natural surface properties, whereas the near-infrared to shortwave infrared bands effectively highlight vegetation physiological states and material differences in artificial structures. Consequently, the model tends to assign higher weights to these bands to emphasize key discriminative information. These results strongly validate the effectiveness of the SSBC-LFT model in spectral band selection for complex RS scenarios, enabling precise information extraction, suppression of redundancy, and enhanced clustering accuracy and robustness.

L. Convergence Analysis

The proposed SSBC-LFT model fundamentally relies on minimizing (13) to achieve optimal clustering results. To evaluate the convergence of the model, we analyze the changes in (13) over different iterations. Fig. 10 illustrates the curves of the objective function values versus the number of iterations on the multimodal RS datasets: Trento, MUUFL, and Houston. As shown in the figure, the objective function values stabilize within 20 iterations across all datasets. These experimental results clearly demonstrate the fast convergence and stability of the proposed model.

VI. Conclusion

Our proposed SSBC-LFT model introduces innovative elements that significantly enhance the exploitation of spatial structures, the scalability of multimodal data processing, and the robustness of clustering performance under noisy conditions in hyperspectral and LiDAR data analysis. By effectively integrating spatially enriched bipartite graph construction, adaptive spectral band selection, and the LFTNN, our approach consistently achieves superior clustering performance, as demonstrated by increased OA and Kappa compared to state-of-the-art methods. The bipartite graph meticulously incorporates spatial context, thereby improving precision, while spectral band selection optimally refines features, significantly reducing dimensionality. Furthermore, the LFTNN efficiently mitigates high-frequency noise, leading to enhanced robustness and boosting overall classification accuracy. Collectively, SSBC-LFT substantially advances multimodal RS clustering, addressing complex challenges in large-scale hyperspectral data analysis more efficiently and more effectively.

VII. Limitations and Future Directions

Despite SSBC-LFT's advancements, our framework has several limitations that require further investigation.

1) *Multiscale Superpixel Segmentation:* The proposed SSBC-LFT model employs a single-scale ERS algorithm to generate superpixels for spatial anchor construction. In future work, a multiscale superpixel segmentation mechanism could be introduced to further enhance feature representation. Specifically, superpixel partitioning can be performed at coarse, medium, and fine scales, with spatial bipartite graphs independently constructed at each scale. This strategy not only preserves the global structural integrity of the majority classes but also enhances the representational independence and discriminability of the minority classes at finer granularities.
2) *Fixed Low-Frequency Norm Constraints:* Although the proposed LFTNN regularization is effective in suppressing high-frequency noise, its excessive application may lead to the over-smoothing of feature representations. To address this limitation, a promising future direction is to develop a structure-aware dynamic frequency filtering strategy. Specifically, this strategy dynamically evaluates the local structural complexity within each homogeneous region and adaptively adjusts the strength of frequency-domain filtering accordingly, thereby enhancing denoising performance while effectively preserving boundary structures and fine-grained details.
3) *Modality-Specific Architecture:* While demonstrating effectiveness in hyperspectral–LiDAR fusion, our method contains modality-specific components that may limit direct extension to emerging sensors (e.g., SAR, thermal, or multitemporal acquisitions). Future work should aim to establish a unified framework for cross-modal feature representation that can effectively integrate diverse sensing modalities, particularly addressing phase-sensitive LiDAR techniques through complex-domain extensions.

These identified limitations simultaneously define promising research trajectories. Overcoming these challenges will enhance SSBC-LFT's theoretical foundation while expanding its applicability in next-generation multimodal RS systems.

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